

Section 1: Anaconda and Jupyter o Section 2: I/O Problems oooooooooooo Section 3: 2D plots ooo Section 4: Anonymity ooooooo Introduction to Python for Mathematics A Crash Course Simon Shaw Mathematics August 19, 2026 © Simon Shaw (2026) Section 1: Anaconda and Jupyter • Section 2: I/O Problems oooooooooooo Section 3: 2D plots ooo Section 4: Anonymity ooooooo Getting started - Log In - Start Anaconda Navigator (use search box) - untick box, click OK, or whatever you choose. - Press 'LAUNCH' for jupyter notebook (not jupyter lab) - Choose 'python 3' from 'new' menu on right - In the first cell type 2+3 followed by SHIFT-RETURN. Did you get 5?	Section 1: Anaconda and Jupyter o Section 2: I/O Problems oooooooooooo Section 3: 2D plots ooo Section 4: Anonymity ooooooo Contents - Anaconda and Jupyter - Leontief Input-Output Problems in Economics 2D Plotting - Anonymity
© Simon Shaw (2026) Section 1: Anaconda and Jupyter o Section 2: I/O Problems •oooooooooooo Section 3: 2D plots ooo Section 4: Anonymity ooooooo Input Output Problems in Economics Consider the tourist economy in a small resort. The major industries are A: Accommodation — rentals, hotels, B & B's, ... F: Food & Drink — restaurants, kiosks, pubs, take away, ... E: Entertainment — theatre, cinema, nightclubs, ... T: Transportation — buses, trains, taxis, ferries, ... The turnover of each of these industries will contain cash inputs from themselves and the others, as well as from external demand like tourists, other industrial and commercial sectors, etc. Reference Mathematics for Economics and Business, Ian Jacques, Prentice Hall, 4 ed. 2003.	© Simon Shaw (2026) Section 1: Anaconda and Jupyter o Section 2: I/O Problems ••oooooooooooo Section 3: 2D plots ooo Section 4: Anonymity ooooooo Let's consider just A and F. Suppose that ... Each £1 of A's turnover requires an input of 10p of its own turnover plus 50p of F's. Each £1 of F's turnover requires an input of 20p of its own turnover plus 30p of A's. So, assuming these proportions are constant across all turnover levels, if we want A to turn over £50,000 and F to turnover £40,000 then: For A: £50,000 requires £5000 (0.1 of £50,000) from itself, plus £20,000 (0.5 of £40,000) from F. For F: £40,000 requires £15,000 (0.3 of £50,000) from A, plus £8000 (0.2 of £40,000) from itself. Get the idea? Easily generalised to more industries. Look carefully: You can see matrices at work here. Let's figure it out. ...
© Simon Shaw (2026) Section 1: Anaconda and Jupyter o Section 2: I/O Problems ooo•oooooooooooo Section 3: 2D plots ooo Section 4: Anonymity ooooooo Look at it mathematically. ... Each £1 of A's turnover requires an input of 10p of its own turnover plus 50p of F's. Each £1 of F's turnover requires an input of 20p of its own turnover plus 30p of A's. Let x_1 denote A's turnover and x_2 denote F's turnover. Also d_1 (resp. d_2) denote external demand for A (resp. F). Then: $\begin{aligned} x_1 &= \overbrace{0.1x_1 + 0.5x_2 + d_1}^{\text{payments to A}} \\ x_2 &= \overbrace{0.3x_1 + 0.2x_2 + d_2}^{\text{payments to F}} \end{aligned}$ Can you see the matrix now?	© Simon Shaw (2026) Section 1: Anaconda and Jupyter o Section 2: I/O Problems oooo•oooooooooooo Section 3: 2D plots ooo Section 4: Anonymity ooooooo Leontief's input-output model Let x_1 denote A's turnover and x_2 denote F's turnover. Let d_1 (resp. d_2) denote external demand for A (resp. F). Then: $\begin{aligned} x_1 &= 0.1x_1 + 0.5x_2 + d_1 \\ x_2 &= 0.3x_1 + 0.2x_2 + d_2 \end{aligned}$ This can be written as $\mathbf{x} = \mathbf{Ax} + \mathbf{d}$ for $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad \mathbf{d} = \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} \quad \text{and} \quad \mathbf{A} = \begin{pmatrix} 0.1 & 0.5 \\ 0.3 & 0.2 \end{pmatrix}.$ In this model $\mathbf{A}$ is called the matrix of technical coefficients, or the technology matrix. (Developed by Wassily Leontief.) $\mathbf{A}$'s columns give the inputs needed for one unit of output.

Section 1: Anaconda and Jupyter 0 Section 2: I/O Problems 00000●00000 Section 3: 2D plots 000 Section 4: Anonymity 0000000 Using the model The Leontief input-output model $x = Ax + d$ can be used to answer three key questions: Given total output x, how much is available for demand d? Find $d = (I - A)x$ How much total output x is required to satisfy a given level of demand d? Solve $(I - A)x = d$ How should output x change if demand changes by Δd? Solve $(I - A)\Delta x = \Delta d$ We'll do these by hand first, and then with python. Remember that $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{\det} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ © Simon Shaw (2026) Section 1: Anaconda and Jupyter 0 Section 2: I/O Problems 00000000000● Section 3: 2D plots 000 Section 4: Anonymity 00000000	Section 1: Anaconda and Jupyter 0 Section 2: I/O Problems 000000●0000 Section 3: 2D plots 000 Section 4: Anonymity 0000000 Use $x = Ax + d$ with $A = \begin{pmatrix} 0.1 & 0.5 \\ 0.3 & 0.2 \end{pmatrix}$ to answer these: Given total output $x = \begin{pmatrix} 50000 \\ 40000 \end{pmatrix}$, how much is available for demand d? Find $d = (I - A)x = \begin{pmatrix} 0.9 & -0.5 \\ -0.3 & 0.8 \end{pmatrix} \begin{pmatrix} 50000 \\ 40000 \end{pmatrix} = \begin{pmatrix} 25000 \\ 17000 \end{pmatrix}$ How much total output x is required to satisfy a given level of demand $d = \begin{pmatrix} 35000 \\ 29000 \end{pmatrix}$? Solve $(I - A)x = d \implies x = \frac{10}{57} \begin{pmatrix} 8 & 5 \\ 3 & 9 \end{pmatrix} \begin{pmatrix} 35000 \\ 29000 \end{pmatrix} \approx \begin{pmatrix} 74561 \\ 64210 \end{pmatrix}$ How should output x change if demand changes by $\Delta d = \begin{pmatrix} 2500 \\ 1900 \end{pmatrix}$? Solve $(I - A)\Delta x = \Delta d \implies \Delta x = \frac{10}{57} \begin{pmatrix} 8 & 5 \\ 3 & 9 \end{pmatrix} \begin{pmatrix} 2500 \\ 1900 \end{pmatrix} \approx \begin{pmatrix} 5175 \\ 4315 \end{pmatrix}$ Now let's do this in python. . . © Simon Shaw (2026) Section 1: Anaconda and Jupyter 0 Section 2: I/O Problems 00000000000● Section 3: 2D plots 000 Section 4: Anonymity 0000000
Section 1: Anaconda and Jupyter 0 Section 2: I/O Problems 00000000000● Section 3: 2D plots 000 Section 4: Anonymity 00000000 The key commands <pre>import numpy as np A = np.array([[0.1, 0.5],[0.3, 0.2]]) Id = np.eye(2) d = np.array([[35000],[29000]]) print(np.linalg.solve(Id-A, d)) Dd = np.array([[2500],[1900]]) print(np.linalg.solve(Id-A, Dd))</pre> # import numerical python # our technology matrix # 2 by 2 identity matrix # demand vector d # solve (I-A) x = d for x # change in demand vector, Dd # solve (I-A) Dx = Dd for Dx Note you can start a new cell whenever you like. Do so frequently. © Simon Shaw (2026) Section 1: Anaconda and Jupyter 0 Section 2: I/O Problems 00000000000● Section 3: 2D plots 000 Section 4: Anonymity 00000000	Section 1: Anaconda and Jupyter 0 Section 2: I/O Problems 00000000000● Section 3: 2D plots 000 Section 4: Anonymity 0000000 Exercise - use python Given the technology matrix $A = \begin{pmatrix} 0.3 & 0.2 \\ 0.1 & 0.6 \end{pmatrix}$. . . Given total output $x = \begin{pmatrix} 70000 \\ 90000 \end{pmatrix}$, how much is available for demand d? Find $d = (I - A)x$ How much total output x is required to satisfy a given level of demand $d = \begin{pmatrix} 52000 \\ 48000 \end{pmatrix}$? Solve $(I - A)x = d$ How should output x change if demand changes by $\Delta d = \begin{pmatrix} 5200 \\ 4100 \end{pmatrix}$? Solve $(I - A)\Delta x = \Delta d$ © Simon Shaw (2026) Section 1: Anaconda and Jupyter 0 Section 2: I/O Problems 00000000000● Section 3: 2D plots 000 Section 4: Anonymity 0000000
Section 1: Anaconda and Jupyter 0 Section 2: I/O Problems 00000000000● Section 3: 2D plots 000 Section 4: Anonymity 00000000 Exercise We started with a tourist economy in a small resort with the major industries: A (Accommodation), F (Food & Drink), E (Entertainment) and T (Transportation). The technology matrix for this economy is $A = \begin{pmatrix} 0.15 & 0.12 & 0.05 & 0.03 \\ 0.17 & 0.16 & 0.04 & 0.04 \\ 0.03 & 0.08 & 0.18 & 0.22 \\ 0.07 & 0.18 & 0.03 & 0.19 \end{pmatrix}$ How much is left for demand d with a total output $x = \begin{pmatrix} 89000 & 55000 & 47000 & 76000 \end{pmatrix}^T$? How much total output x is needed for demand $d = \begin{pmatrix} 55000 & 24000 & 18000 & 40000 \end{pmatrix}^T$? How should output x change if demand changes by $\Delta d = \begin{pmatrix} -5000 & 350 & 2300 & -500 \end{pmatrix}^T$? © Simon Shaw (2026) Section 1: Anaconda and Jupyter 0 Section 2: I/O Problems 00000000000● Section 3: 2D plots ●00 Section 4: Anonymity 00000000	Section 1: Anaconda and Jupyter 0 Section 2: I/O Problems 00000000000● Section 3: 2D plots 000 Section 4: Anonymity 0000000 Finishing up There is much much much more to python, and numpy. The notebook contains some eigenvalue and SVD examples We're just scratching the surface As is the nature of a crash course Let's look now at how to plot graphs © Simon Shaw (2026) Section 1: Anaconda and Jupyter 0 Section 2: I/O Problems 00000000000● Section 3: 2D plots ●00 Section 4: Anonymity 0000000
Section 1: Anaconda and Jupyter 0 Section 2: I/O Problems 00000000000● Section 3: 2D plots ●00 Section 4: Anonymity 00000000 2D Plotting © Simon Shaw (2026) Section 1: Anaconda and Jupyter 0 Section 2: I/O Problems 00000000000● Section 3: 2D plots ●00 Section 4: Anonymity 0000000	Section 1: Anaconda and Jupyter 0 Section 2: I/O Problems 00000000000● Section 3: 2D plots ●00 Section 4: Anonymity 0000000 Plotting in 2D We jump straight in. Plot $\cos(2\pi x)$ in solid blue and $\exp(\sin(4\pi x))$ in dashed red for $x \in [-1, 5]$ <pre>import matplotlib.pyplot as plt import numpy as np x = np.arange(-1,5,0.01) y1, y2 = np.cos(2*np.pi*x), np.exp(np.sin(4*np.pi*x)) plt.plot(x,y1, 'b-') plt.plot(x,y2, 'r--') plt.axis([-2, 6, -2, 4]) plt.legend(['cos', 'exp(sin)']) plt.xlabel(r'\$x_1\$'); plt.ylabel('\$y_1\$ and \$y_2\$') plt.savefig('my2Dplot.png', dpi=600) plt.savefig('my2Dplot.eps', dpi=600)</pre> 4 — cos - - exp(sin) © Simon Shaw (2026) Section 1: Anaconda and Jupyter 0 Section 2: I/O Problems 00000000000● Section 3: 2D plots ●00 Section 4: Anonymity 0000000

Section 1: Anaconda and Jupyter
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Section 2: I/O Problems
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Section 3: 2D plots
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Section 4: Anonymity
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Exercise

In python...

Plot $2^3 \sin(3\pi x)$ in solid dash-dot blue and $\ln(1.2 + \sin(3\pi x))$ in dotted red for $x \in [-4, 3]$

Hint: for the line-styles use

```
plt.plot(x,y1, 'b-.'')
plt.plot(x,y2, 'r:')
```

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Section 1: Anaconda and Jupyter
o

Section 2: I/O Problems
oooooooooooo

Section 3: 2D plots
ooo

Section 4: Anonymity
●ooooo

How anonymous is anonymized data?

A university collects answers to personal questions from all of its students.

Each student's answer has their name, date of birth, gender and department.

On average a department has 250 students in each year.

We assume the UK setup where students attend for three years.

The names are erased: how anonymous are the resulting data?

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oSection 2: I/O Problems
ooooooooooooSection 3: 2D plots
oooSection 4: Anonymity
●ooooo

How Anonymous is Anonymised Data?

Here's part of the dataset

NAME	D.O.B.	GENDER	DEPARTMENT
:	:	:	:
Ringo Starr	26/07/43	M	Music
Al Gebra	12/08/05	F	Maths
Sandie Shaw	16/02/38	F	Puppetry
Michael Mouse	17/04/92	F	Computing
Mr Pink	4/12/56	M	Criminology
L.O. Gear	11/9/23	F	Engineering
Donkey Kong	23/10/73	M	Video Games
:	:	:	:

The names are erased — can one line identify the person? Is this enough information to identify the person?

How anonymous is a dataset like this?

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oSection 2: I/O Problems
ooooooooooooSection 3: 2D plots
oooSection 4: Anonymity
●ooooo

Let's simulate

We assume that all students are born within a three year window, and that each department has 750 students across its three years.

- There are $365 \times 3 = 1095$ possible birthdays
- For each there are at least 2 possible genders
- So, for a given department, there are $d = 1095 \times 2 = 2190$ possible entries among $N = 750$ students.
- Can you see why anonymity might not be assured?

Think about a line of $N = 2190$ empty buckets. Now throw $N = 750$ balls at random into the buckets.

Most will stay empty. Some will have just one ball — the 'loners'.

The proportion of N having just one ball estimates the probability that line of anonymised data occurs just once in the department.

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Section 1: Anaconda and Jupyter
oSection 2: I/O Problems
ooooooooooooSection 3: 2D plots
oooSection 4: Anonymity
●ooooo

Background and Exercise

- There is about a 70% probability that a student can be identified from this anonymised data.
- A 'near exact' solution is $\exp(-N/d) \approx 71\%$.

My main reference is John D Cook:

www.johndcook.com/blog/2018/12/07/simulating-zipcode-sex-birthdate/Which itself references the paper *Only You, Your Doctor, and Many Others May Know*, Latanya Sweeney:<https://techscience.org/a/2015092903/>

Exercise: What is the probability that a line of anonymised data can be narrowed down to at most two students? Or three? Or four?

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Section 1: Anaconda and Jupyter
oSection 2: I/O Problems
ooooooooooooSection 3: 2D plots
oooSection 4: Anonymity
●ooooo

Simulation — Planning the code

We are going to generate a list of $d = 2190$ zeros.We'll then generate $N = 750$ random integers $z \in \{1, 2, \dots, 2190\}$.For each z we'll add one to the z^{th} item in the list, d .In the end, the n^{th} item in the list, d , tells us how many students share that same data.

We want to find the 'loners' — the buckets with only one item in them.

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Section 1: Anaconda and Jupyter
oSection 2: I/O Problems
ooooooooooooSection 3: 2D plots
oooSection 4: Anonymity
●ooooo

The End

That's it!

Thanks for listening

There's lots more to learn — as ever!

There's a few extra things in the notebook

Good Luck!

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Section 1: Anaconda and Jupyter
oSection 2: I/O Problems
ooooooooooooSection 3: 2D plots
oooSection 4: Anonymity
●ooooo